

Technical Documentation

Project: Your Library

Authors:

Emilia Lehmann

Wiktoria Kowalska

Weronika Kościug

Project Purpose

Purpose of the project is to develop a web application that enables users to manage their personal book collection and share books with other members of the application. This application was designed to simplify the organization of personal libraries, you can add books already read, currently reading, or planned to read in the future. You can store there all variety of information about each position in your library. What is more, browse your friends, view their collections, share opinions and leave reviews which you can also look through. The system also provides borrowing mechanism between friends. In one place you can track both borrowed and lent books in dedicated sections of the application.

System Architecture

The application follows the Model-View-Controller (MVC) architectural pattern provided by ASP.NET Core

Models

Represent the application's domain entities and database objects

Views

Provide the user interface using HTML, CSS, JavaScript and Razor Views

Controllers

Contain the application logic and handle communication between views and the database

Entity Framework Core is used as the Object-Relational-Mapper (ORM) to manage database operations and relationships between entities. User authentication and authorization are implemented using ASP.NET Identity.

The application also integrates external services, including Google Books API for retrieving book information, and Gmail SMTP for password recovery functionality.

Database Structure:

The database is implemented using Microsoft SQL Server and Entity Framework Core. Database is responsible for storing user accounts, personal book collections, reviews, friendships and book borrowing transactions between users.

1. There are six main entities:

ApplicationUser (AspNetUsers)

Stores information about registered users, their authentication data, profile details. Has relationships with books, borrow transactions and friends.

Most important fields: Id, UserName, Email, PasswordHash, ShelfVisibility, DisplayName, Avatar, AvatarImagePath

Books

Has general information about books including title, author, genre, cover image etc. .

Most important fields: BookId, Title, Author, Description, Genre, AgeRating, ImageURL

UserBooks

Represents a specific book added to user's personal library. Stores information such as reading status, notes, storage location, media format and ownership status.

Most important fields: UserBookId, Bookmark, Location, Notes, IsBorrowed, IsOwned, ReadStatus, ApplicationUserId, BookId, ReviewId

Borrows

Stores information about lending and borrowing transactions between users. There are borrow status, borrowing dates, return dates, borrower notes and personal review created during the borrowing process.

Most important fields: BorrowId, StatusBorrow, BorrowDate, ReturnDate, UserBookId, ApplicationUserId

Reviews

Stores user ratings and review comments associated with books in the library.

Most important fields: ReviewId, Rating, ReviewComment

Friendship

Represents connections between users. Stores friend requests, accepted friendships, and date when friendship was created.

Most important fields: FriendshipId, FriendStatus, RequesterId, ReceiverId, CreatedAt

2.Relationships

ApplicationUser (1) --- (N) UserBooks

Books (1) --- (N) UserBooks

ApplicationUser (1) --- (N) Borrows

UserBooks (1) --- (N) Borrows

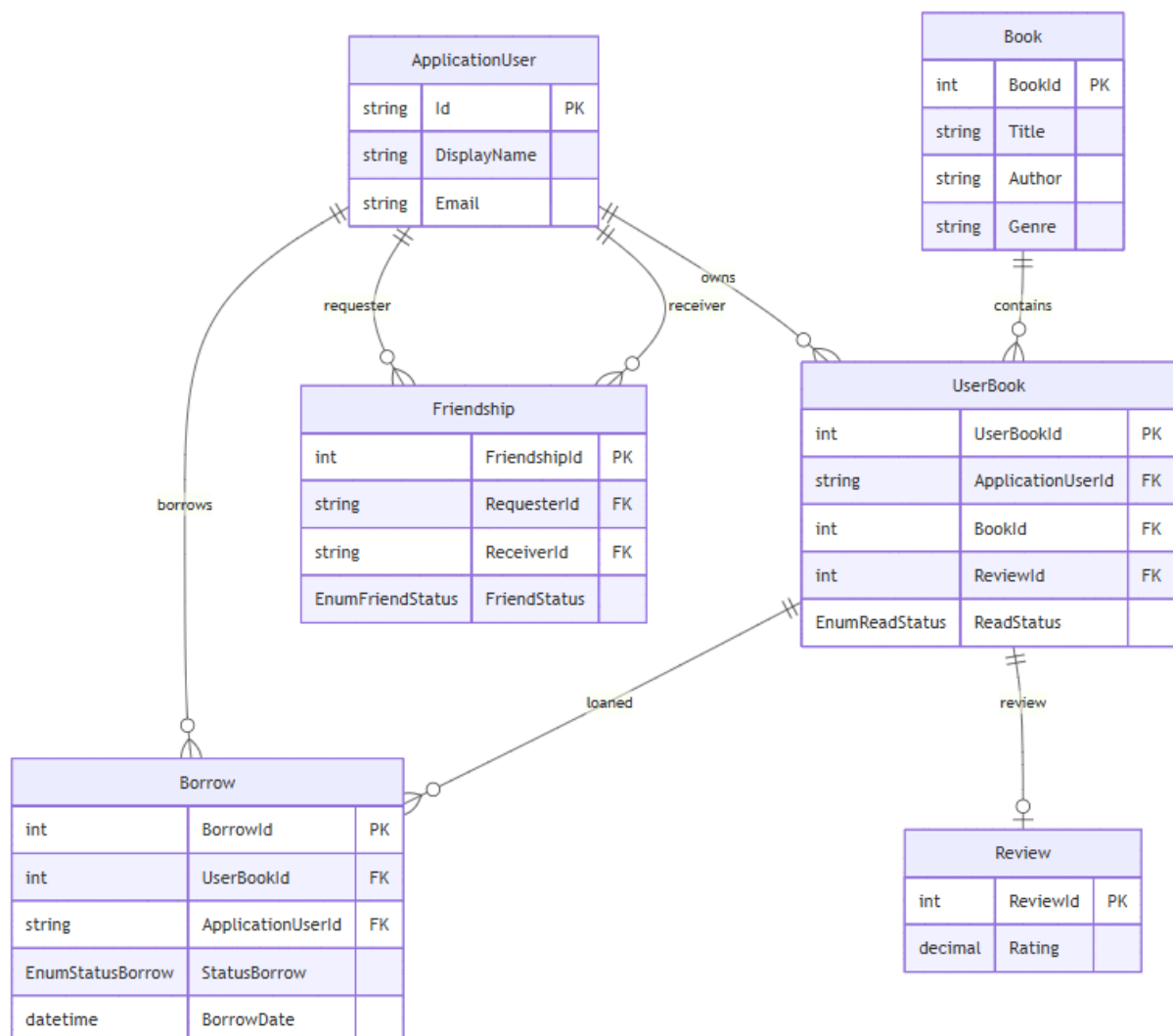
(As Requester) ApplicationUser (1) --- (N) Friendship

(As Receiver) ApplicationUser (1) --- (N) Friendship

UserBooks (1) --- (0..1) Reviews

3.Diagram

There are shown only most important fields in tables to visualize connections and keys between tables.



Main Implemented Functionalities:

The application provides the following core functionalities:

User Management:

User registration and login

Password reset using email

User profile editing

Avatar customization and profile image upload

Personal Library Management :

Adding books to a personal library

Editing and deleting books from your collection

Managing personal notes and storage locations

Friendship system:

Sending friendship invitations

Accepting and rejecting requests

Viewing friends and their collections

Book borrowing system:

Requesting borrows from friends

Tracking borrowed and lent books

Returning borrowed books

Reviews and ratings:

Creating book reviews/ratings

Viewing reviews from other users